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Clarification of the properties and accumulation effects of ion tracks in CeO_2

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Abstract

In order to understand the properties of ion tracks and the microstructural evolution under accumulation of ion tracks and Xe ions in CeO_2, 70–210 MeV $Xe^{10\sim14+}$ ions irradiation examinations and pre-implantations of 240 keV Xe ions have been done at a tandem accelerator facility and an ion implanter facility of JAEA-Tokai. The microstructure observations were performed by means of a transmission electron microscope (TEM) and a scanning electron microscope (SEM) in CRIEPI.

Measurements of the diameter of ion tracks with the irradiation temperature, between room temperature and 800 °C, clarify that the prominent changes of ion tracks are hardly observed up to 800 °C. By cross-sectional observation, it becomes clear that the threshold electronic stopping power of ion track formation is around 15–16 keV/nm in case of Xe ions irradiation. 210 MeV Xe^{14+} ion irradiations cause a surface roughness on CeO_2 in the ion fluence range between 5×10^{14} ions/cm^2 and 1×10^{15} ions/cm^2.
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1. Introduction

For reducing the total amount of spent fuels of light-water reactor (LWR) and eventually the fuel cycle costs, burnup extension of LWR fuels have been promoted step-by-step. At the peripheral region of high burnup fuel pellets, a crystallographic re-structuring is observed, commonly called the "rim structure [1–3]" or "high burnup structure", which is characterized by the existence of highly dense small sub-grains, whose size is approximately 100 nm, and the accumulation of small pores with average size around 1 μm. This restructuring can influence the fuel performance including, e.g. fission gas release, fuel temperature, hardness and swelling. Therefore, in order to extend the burnup of LWR fuels, formation and growth mechanism of the rim structure should be clarified. This structure shall be formed by the accumulation and mutual interactions of radiation damages, fission products (FPs) and electronic excitations deposited partially by nuclear fissions [4]. For separating and evaluating these effects, ion irradiation studies using high-energy ion accelerators and transmission electron microscope (TEM)/scanning electron microscope (SEM) observations after irradiation are effective. Moreover, in order to clarify the irradiation effects of fluorite structure ceramics effectively, CeO_2 have been used as a simulation of UO_2. Some of information about the irradiation effects in CeO_2 and UO_2 has been reported in previous papers [5–8], though; they have not been enough for understanding the formation mechanism of rim structure.

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This paper attempts to clarify (1) the properties of ion tracks, such as the temperature effects and the threshold electronic stopping power (S_e) of ion tracks formation in CeO_2 and (2) the microstructural evolution under accumulation of ion tracks and pre-implantation of Xe ions in CeO_2.

2. Experimental

In this research, 4N CeO_2 powder supplied by RARE METALLIC Co. was pressed and sintered as 8 mmϕ $\times$ 1 mm bulk at 1400 °C for 8 h, then cooling to room temperature for 8 h in air. The sintered bulk specimen was ground and polished for ion irradiation examination. For producing TEM samples, some of the bulk specimen was cut out as the 3 mmϕ discs, dimpled and thinned by ion sputtering of 5 keV Ar^+ ions. A focused ion beam (FIB) system by Hitachi FB-2000A was also adopted for making cross sectional TEM samples [9].

Ion irradiation examinations have been done at JAEA-Tokai Tandem accelerator facility, where an ECR ion source is installed and is able to irradiate high energy inert gases such as Xe, Kr. In this study, irradiation examinations on the samples with 70 – 210 MeV $Xe^{10\sim14+}$ ions were conducted at temperatures from room temperature to 800 °C. Moreover, in order to simulate the retained FP gas in high burnup fuels, 240 keV Xe ions are pre-implanted in CeO_2 by use of an ion implanter with ECR ion source.

Microstructural evolutions in the samples were observed by use of a 300 kV field emission transmission electron microscope (FE-TEM); Hitachi HF-3000 and a field emission scanning electron microscope (FE-SEM); JEOL JSM-6340 F at CRIEPI.

3. Results and discussion

3.1. Properties of ion tracks in CeO_2

For understanding the temperature effects on the configuration of ion tracks, the microstructure and the diameter of ion tracks are measured as a function of irradiation temperature. Fig. 1(a)–(d) indicates the typical bright field images of ion tracks in CeO_2 under irradiation with 210 MeV Xe^{+14} ions to a fluence 1×10^{11} ions/cm^2 at room temperature (a), to a fluence 5×10^{11} ions/cm^2 at 500 °C (b), to a fluence 5×10^{11} ions/cm^2 at 600 °C (c) and to a fluence 5×10^{11} ions/cm^2 at 800 °C (d). These images indicate that the cross sections of ion tracks become blur over 600 °C, though, the ion tracks still exist till 800 °C. Fig. 1(e) shows the mean diameter of ion tracks by 100 MeV and 210 MeV Xe^{+14} ion irradiation as a function of irradiation temperature (°C). This figure clarified that the mean diameter of ion tracks is decreasing as irradiation temperature increases and is decreased by about 23% at 800 °C. These results indicate that the ion tracks can exist up to 800 °C and the temperature range from room temperature to 800 °C does not significantly affect annealing of ion tracks. That suggests that the effects of accumulation of ion tracks on microstructure changes in CeO_2 should be taken into consideration.

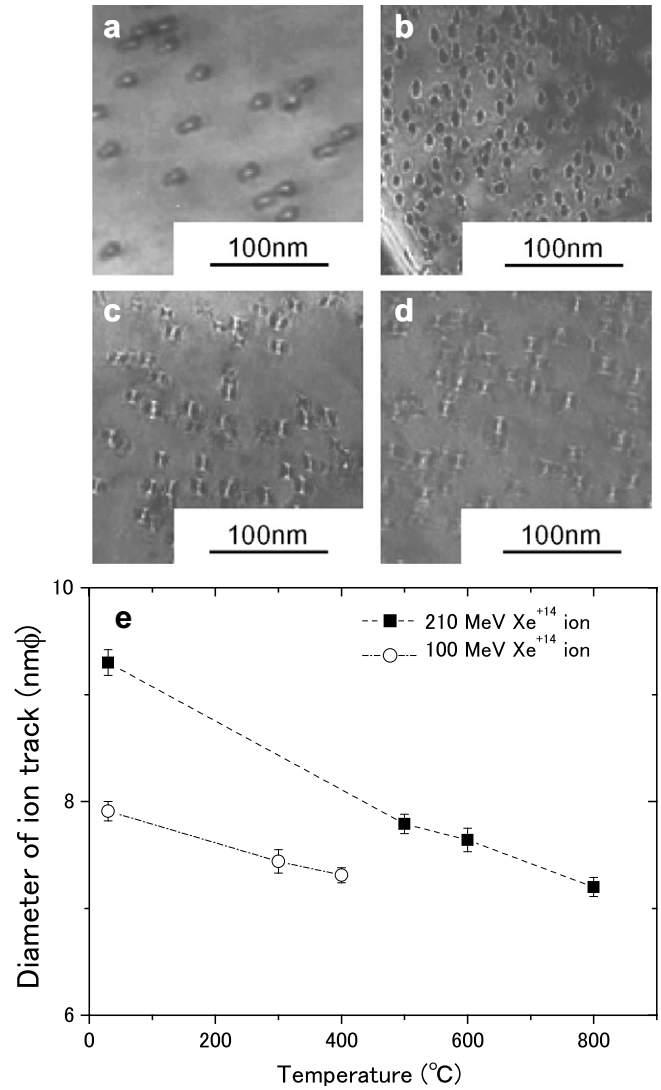

Fig. 1. Typical bright field images of ion tracks under irradiation with 210 MeV Xe^{+14} ions (a–d) and the mean diameter of ion tracks by 100 MeV and 210 MeV Xe^{+14} ion irradiation as a function of irradiation temperature (°C) (e). The irradiation condition of (a)–(d) is to a fluence 1×10^{11} ions/cm^2 at room temperature (a), to a fluence 5×10^{11} ions/cm^2 at 500 °C (b), to a fluence 5×10^{11} ions/cm^2 at 600 °C (c) and to a fluence 5×10^{11} ions/cm^2 at 800 °C (d), respectively.

The affected area of electronic excitation by fissions in CeO_2 has been clarified to be around 5–7 nmϕ [5]. For understanding the "affected volume of high-density electronic excitation" in CeO_2, the length of ion tracks, that is, the threshold electronic stopping power (S_e) for ion track formation in CeO_2 is indispensable. In order to clarify the depth profile of ion tracks in CeO_2, cross sectional TEM observations by use of a FIB facility and FE-TEM were applied. Fig. 2(a) indicates the depth profile of electronic stopping power (S_e) and nuclear stopping power (S_n) of 210 MeV Xe^{+14} ion in CeO_2 that was estimated

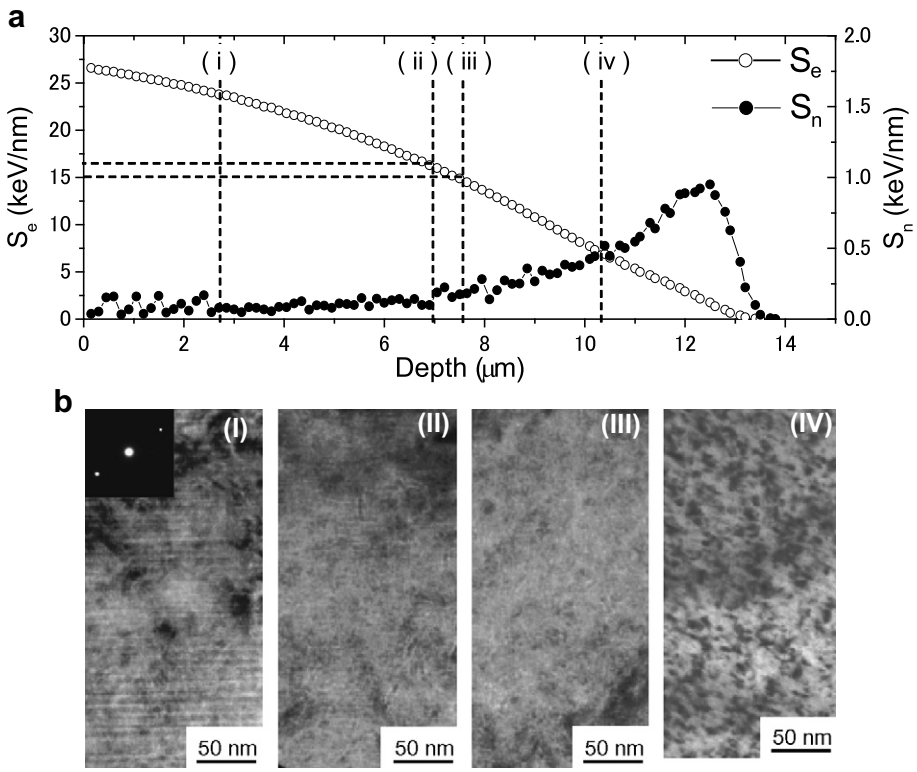

Fig. 2. Depth profile of electronic stopping power (S_e) and nuclear stopping power (S_n) of 210 MeV Xe^{+14} ion in CeO_2 (a) and the typical TEM images in selected depth positions (b). The position of each image is at depth of ~2.7 μm (i), ~7.0 μm (ii), ~7.5 μm (iii) and ~10.3 μm (iv), respectively.

by SRIM 2003 code [10] and Fig. 2(b) shows the typical TEM images at a depth of ~2.7 μm (i), ~7.0 μm (ii), ~7.5 μm (iii) and ~10.3 μm (iv), respectively. At a depth of ~2.7 μm, ion tracks are clearly observed in Fig. 2((b)–(i)). The diameter of ion tracks decreases as the distance from irradiation surface becomes longer and the images of ion tracks become invisible at the depth of around 7.0–7.5 μm, as shown in Fig. 2 ((b)-ii, iii). At the depth of ~10.3 μm, S_e is decreased and S_n become bigger, relatively, high dense defect clusters were observed, as shown in Fig. 2 ((b)-iv). These figures clarify that the depth up to which ion tracks are formed is around 7.0–7.5 μm in case of 210 MeV Xe^{+14} irradiation in CeO_2. This result suggests that the threshold S_e of ion track formation in CeO_2 is around 15–16 keV/nm. In case of 80 MeV Xe^{+14} irradiation, the tracks were observed up to a depth of ~2.1 μm and the threshold S_e of ion track formation in CeO_2 can be estimated ~15 keV/nm. In order to clarify the value with a higher accuracy, cross sectional TEM observations of CeO_2 under irradiation with several irradiation energies will be treated.

3.2. Accumulation effects of ion tracks and pre-implantation of Xe ions

In order to clarify the effects of accumulation of ion tracks on the microstructure change in CeO_2, high fluence irradiations up to 2×10^{15} ions/cm^2 have been done and the irradiated samples have been observed by SEM/TEM. Fig. 3(a) and (b) shows the typical SEM images of irradiated surfaces in CeO_2 under irradiation with 210 MeV Xe^{+14} at 300 °C to a fluence of 3×10^{14} ions/cm^2 (a) and 2×10^{15} ions/cm^2 (b), respectively. This figure indicates that the drastic change of the surface condition is occurred between (a) and (b). High magnification image of the surface roughness in the sample (b), as shown in Fig. 3(c), has similar morphology of small sub-divided grains that are observed in inner surfaces of coarsened bubbles (size: ~1 μm) in high burnup fuels [5]. A cross-sectional TEM image of the part of surface roughness in sample (b) is indicated in Fig. 3(d). In this figure, the grain boundaries are observed at around 100 nm depth from the irradiated surface whose morphology is changed. This result suggests a possibility that the surface roughness is caused by sub-divided grains. Several irradiation examinations whose irradiation temperature up to 800 °C confirm that the drastic change of surface morphology in all irradiated area is caused at a fluence to over 5×10^{14} ions/cm^2. These results clarify that the overlapping of ion tracks is needed for the drastic change of surface morphology.

For the simulation of high burnup fuels, high fluence pre-implantation of 240 keV Xe ions (~2×10^{16} ions/cm^2) at room temperature are prepared before irradiation with 210 MeV Xe ions. The density of pre-implanted Xe ions in the ion range (~50 nm) is estimated as around 2×10^{21} ions/cm^3 and the density corresponds with that of high burnup fuels (~70 MWd/kgU). Fig. 4 indicates the typical SEM image of irradiated surface in CeO_2 under

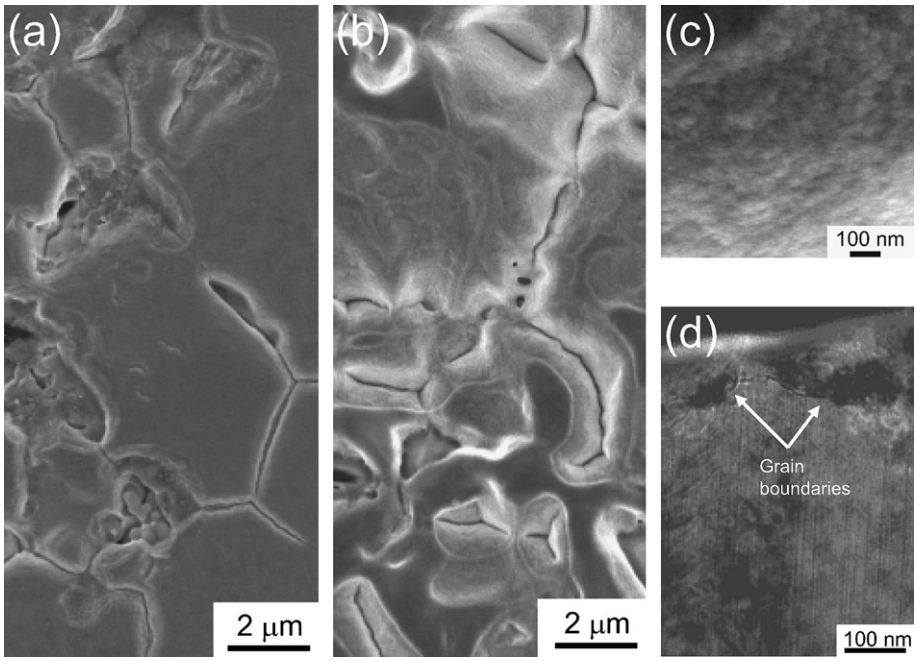

Fig. 3. Typical SEM images of irradiated surfaces in CeO_2 under irradiation with 210 MeV Xe^{+14} at 300 °C to fluences of 3×10^{14} ions/cm^2 (a), 2×10^{15} ions/cm^2 (b); a higher magnification SEM image of sample (b) and (c) and a cross-sectional TEM image for sample (b) and (d), respectively.

irradiation with 210 MeV Xe^{+14} to a fluence of 5×10^{14} ions/cm^2 at 300 °C (a) and multiple irradiation with 240 keV Xe^{+12} to a fluence of 2×10^{16} ions/cm^2 at RT and 210 MeV Xe^{+14} to a fluence of 5×10^{14} ions/cm^2 at 300 °C. This figure suggests that the formation of surface roughness tends to be accelerated by the pre-implantation of Xe ions. In order to understand the formation mechanism of the surface roughness, further multiple ion irradiation examinations and cross-sectional observations will be continued in the near future.

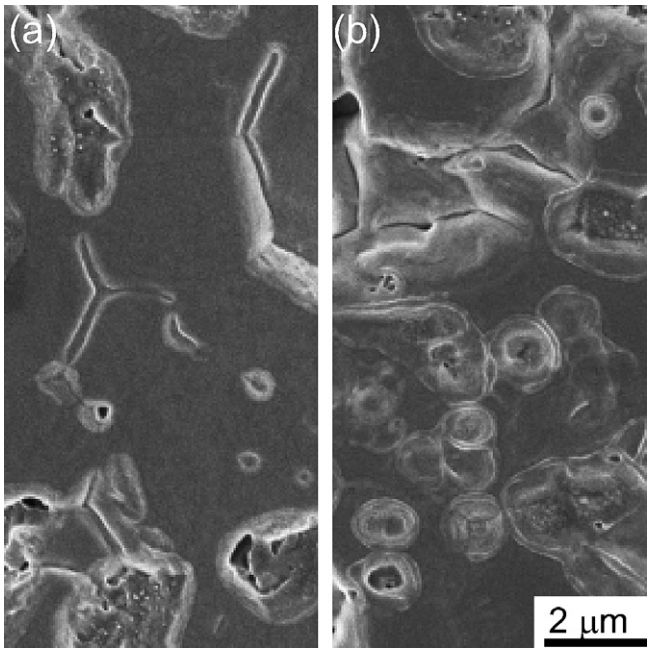

Fig. 4. Typical SEM image of irradiated surface in CeO_2 under irradiation with 210 MeV Xe^{+14} to a fluence of 5×10^{14} ions/cm^2 at 300 °C (a) and multiple irradiation with 240 keV Xe^{+12} to a fluence of 2×10^{16} ions/cm^2 at RT and 210 MeV Xe^{+14} to a fluence of 5×10^{14} ions/cm^2 at 300 °C (b).

4. Conclusions

Multiple ion irradiation examinations with 210 MeV and 240 keV Xe ions on CeO_2 allowed clarifying the properties of ion tracks and the accumulation effects of ion tracks and pre-implantation of Xe ions on the microstructure changes in CeO_2. The main conclusions are,

(1) Observations and diameter measurements of ion tracks in CeO_2 clarified that the ion tracks can exist up to 800 °C and the temperature range from room temperature to 800 °C does not significantly affect annealing of ion tracks. Cross-sectional observation clarified that the threshold electronic stopping power of ion track formation is around 15–16 keV/nm in case of Xe irradiation.
(2) Ion track accumulation cause a surface roughness on CeO_2 in the ion fluence range between 5×10^{14} ions/cm^2 and 1×10^{15} ions/cm^2. High fluence Xe pre-implantation before high energy Xe irradiation tends to accelerate the formation of surface roughness in CeO_2.

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